

Math 170S - Homework 8 - due March 12

Please fill out the teaching evaluations! 1% extra credit will be added to your final grade for the completion of the evaluation form. Thanks in advance for any detailed feedback provided.

Problem 1. 9.1-2

Problem 2. 9.1-6

Hint: Note that H_0 is X is $b(3, 0.3)$, so both parameters of the binomial distribution are given in advance and the correction in the number of degrees of freedom is not needed)

Problem 3. 9.1-11 in the red book/9.1-13 in the blue book

Problem 4. 9.2-3

Problem 5. 9.2-4

Problem 6. 9.2-13 only (c) + find p -value for this test

Problem 7. (based on 9-3-11, Hogg) The driver of a diesel-powered automobile decided to test the quality of three types of diesel fuel sold in the area (whether three means are equal). The test is to be based on miles per gallon, mpg.

- a) Make the usual assumptions (state them explicitly), give a critical region for an $\alpha = 0.05$ significance level.
- b) Construct an ANOVA table and state your conclusion, using the following data:
Brand A: 38.7 39.2 40.1 38.9
Brand B: 41.9 42.3 41.3
Brand C: 40.8 41.2 39.5 38.9 40.3
- c) Give bounds on the p -value for this test.